12.11 Stamps, Cups and e

WORKSHEET

The question

STEP I 1996, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s1-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself For $N = 4$, what must happen? With what probability?

STAGE 1 The distribution

Covers: the distribution of N

Ask yourself For $N = k$, Jim must guess right with $2, 3, \ldots, k-1$ cups and wrong with k cups. Multiply the probabilities.

STAGE 2 The expectation

Covers: the expectation

Pause and try Simplify $k \cdot \frac{k-1}{k!}$. Which famous series do you get?

STAGE 3 The variance

Covers: the variance

Ask yourself Find $\mathrm{E}(N^2)$ by writing $k^2 = k(k-1) + k$, or find $\mathrm{E}\big(N(N-1)\big)$ first.

What's really going on

The same distribution, in disguise

Choose random numbers $U_1, U_2, \ldots$ uniformly from $[0,1]$, and let M be the number needed for their sum to exceed 1. The probability that the first k numbers add up to less than 1 is the volume of the region $\{u_1 + \cdots + u_k < 1, u_i \geq 0\}$, which is $\frac{1}{k!}$ (a triangle of area $\frac{1}{2}$, a tetrahedron of volume $\frac{1}{6}$, and so on). So

$$\mathrm{P}(M > k) = \frac{1}{k!}, \qquad \mathrm{P}(M = k) = \frac{1}{(k-1)!} - \frac{1}{k!} = \frac{k-1}{k!},$$

exactly the distribution of N. In both problems $\mathrm{P}(N > k) = \frac{1}{k!}$: for the cups, that is the probability of $k-1$ correct guesses in a row. Using $\mathrm{E}(N) = \sum_{k \geq 0} \mathrm{P}(N > k)$ (a formula from Section 9.9) gives $\mathrm{E}(N) = \sum \frac{1}{k!} = e$ in one line.

Why e keeps appearing

Whenever a probability involves $\frac{1}{k!}$, whether from k objects in a particular order, k successive events with probabilities $\frac{1}{2}, \frac{1}{3}, \ldots$, or the volume of a simplex, sums over k produce the exponential series. Derangements (Section 9.8) gave $\sum \frac{(-1)^k}{k!} = e^{-1}$; here we have $\sum \frac{1}{k!} = e$. And limits of the form $\left(1 - \frac{1}{n}\right)^n \to e^{-1}$, which come from many independent unlikely events, explain e in the secretary problem and the Poisson distribution.

Monte Carlo e

The "sum exceeds 1" problem gives a way to estimate e by experiment: generate random numbers, count how many are needed to exceed 1, repeat many times and average. It is a cousin of Buffon's needle (Section 12.20), which estimates π. Such *Monte Carlo* methods, named after the casino, use random sampling to estimate quantities that are hard to compute directly; they are now used throughout science, from particle physics to finance, mostly for integrals in many dimensions.

Try this next

1. **The generating function.** Show that $\mathrm{E}(t^N) = 1 + (t-1)e^t$, and use it to find $\mathrm{E}(N)$ and $\mathrm{E}(N(N-1))$.

2. **The tail formula.** Prove that $\mathrm{E}(N) = \sum_{k \geq 0} \mathrm{P}(N > k)$ for a random variable taking non-negative integer values, and use it to find $\mathrm{E}(N)$ in one line.

3. **Starting with one cup.** If the game starts with one cup (which Jim cannot get wrong), how does the distribution change? What is the expected price?

4. **The simplex.** Show that the volume of $\{u_1 + u_2 + u_3 < 1, u_i \geq 0\}$ is $\frac{1}{6}$, and use induction to show the k-dimensional version is $\frac{1}{k!}$.

Further reading

- **Eli Maor, *e: The Story of a Number*** (Princeton University Press). The history of e, including its appearances in probability.
- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), on generating functions.
- **Paul Nahin, *Digital Dice*** (Princeton University Press). Probability puzzles solved by Monte Carlo simulation.

The full guided solution is in Section 12.11:
https://beyondstep.keithmcnulty.org/ch12/stamps-and-e.html